Zelvaritin Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-341/A

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Section 3.2.P.4 — Excipients

For the commercial formulation F-341/A, Primojel (Sodium starch glycolate, disintegrant) is used.

Grade selection and control follow the principles of ICH Q3D.

Formulation Development Rationale

The basis for selection of Primojel is reduced sticking and picking observed during tableting trials, demonstrated across three pilot-scale batches.

Commercial Control Strategy — Excipients

The current approved commercial specification (SPEC-EX-7528 Rev 09) lists Sodium starch glycolate as Primojel.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Cleaning verification swab results were below the health-based exposure limits derived for the product. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Deviations observed during the campaign were investigated and closed with no impact to product quality. No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Cleaning verification swab results were below the health-based exposure limits derived for the product. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development.